



Welcome to

Cloud Computing



2026

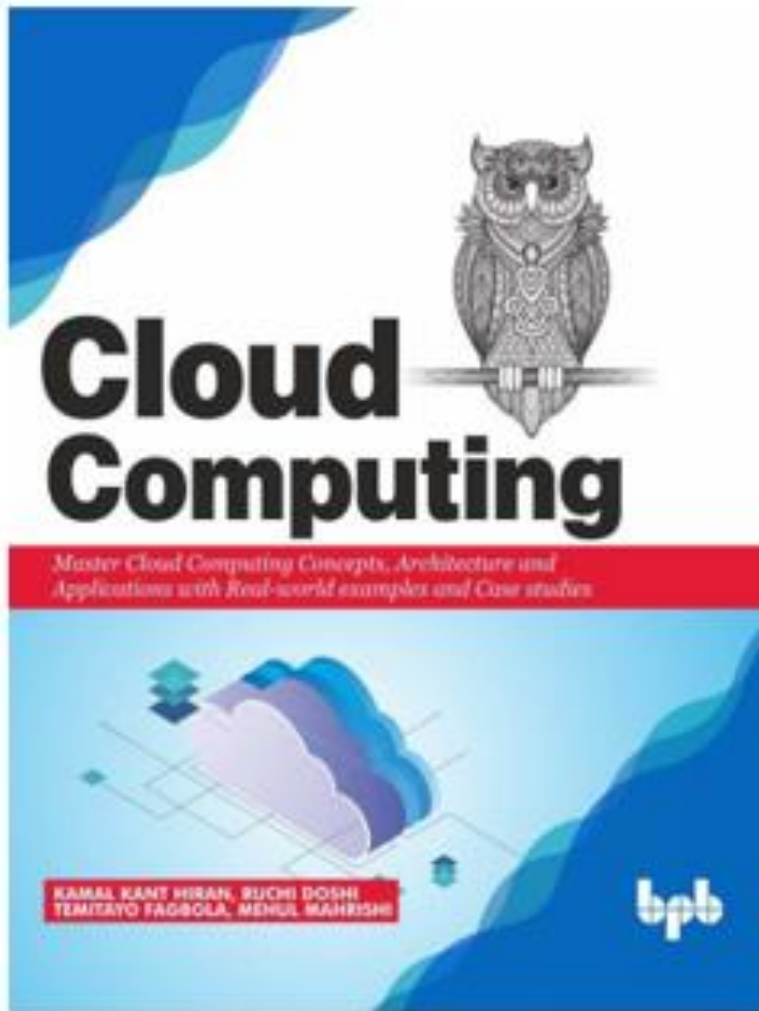
TEXTBOOKS

PRESCRIBED

- **Cloud Computing: Master the Concepts, Architecture and Applications with Real-world Examples and Case Studies**, 1st edition, Ruchi Doshi; Temitayo Fagbola; Mehul Mahrishi , ISBN: 978-93-88511-407
- **Cloud Computing**. Edition: N/A, Kumar, N. S. and Bhatt, P. C. P. ISBN: 978-3-319-77838-9

RECOMMENDED

- **Cloud Computing Applications and Techniques for E-Commerce**, Edition: N/A, Author: Gochhait, S.; Shou, D. T. and Fazalbhoy, S. Year of Publication: 2019



CLOUD COMPUTING:
Master the Concepts, Architecture, and Applications
with Real-world examples and Case studies

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Lesson 1: **Cloud Computing Ecosystem Fundamentals**

Content



CLOUD
COMPUTING
ECOSYSTEM
(CCE)



CLOUD
COMPUTING
CHARACTERISTICS



BUSINESS
DRIVERS OF
CLOUD
COMPUTING
ADOPTION



SUMMARY



Q&A

"**Cloud computing** is a spectrum of things complementing one another and building on a **foundation of sharing**."

Chris Howard, Research Vice President at Gartner

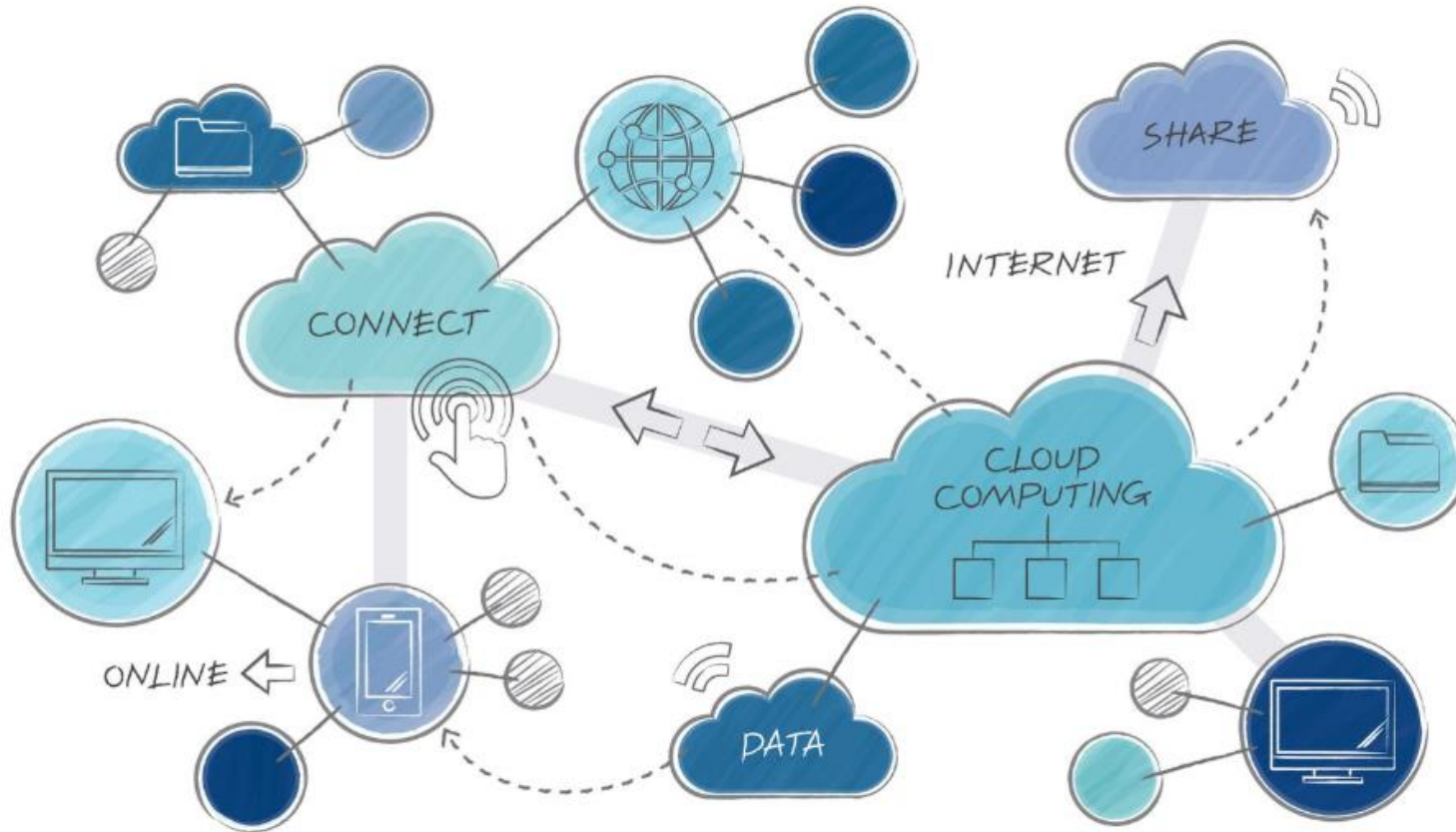


Cloud Computing

The **delivery of computing as a service over a network, whereby distributed resources are provided to the end user as a utility**



CLOUD COMPUTING ECOSYSTEM (CCE)



CLOUD COMPUTING ECOSYSTEM (CCE)

- Is a **dynamic** and **complex** **community** of the **cloud** **computing system** **components** and **the** **stakeholders**

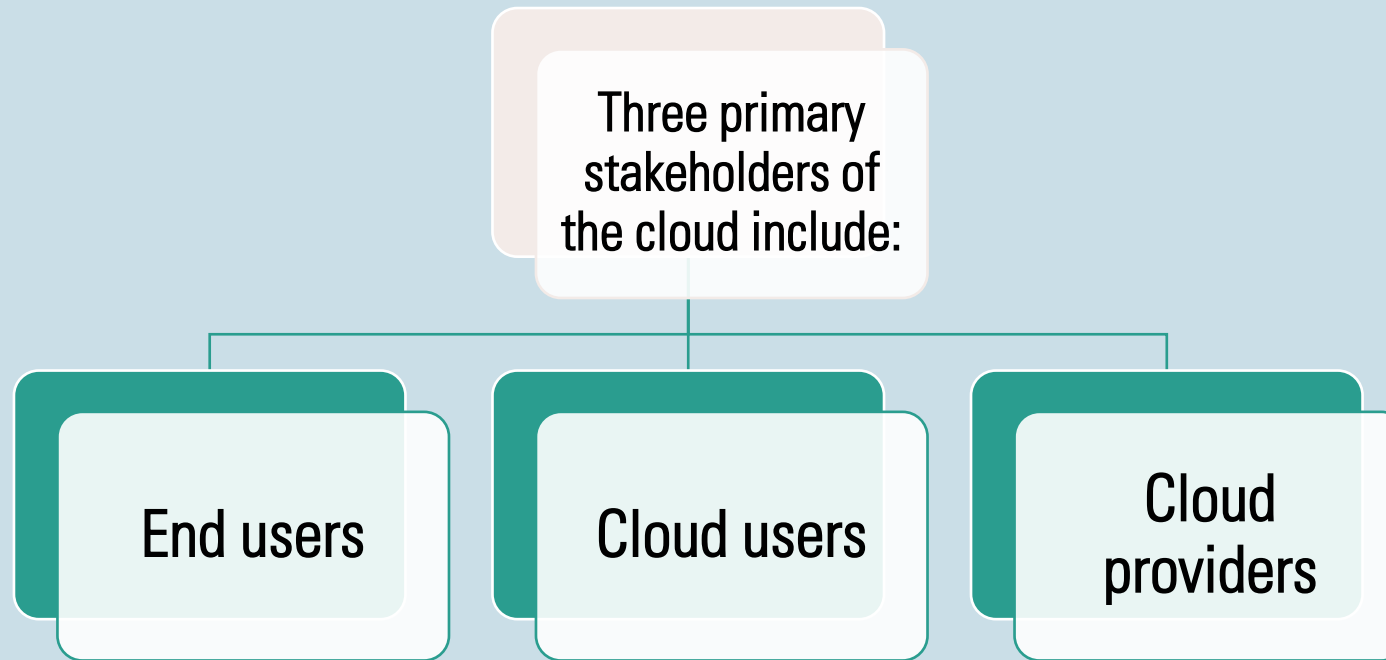
Interdependent components of the CCE include

- Cloud **consultants**
- Cloud **service** providers
- Cloud **end-users** (customers)
- Cloud **product** manufacturers
- Cloud **engineers**
- Cloud **partners**

High-speed network

- Cloud management environment as well as the cloud computing infrastructures and IT resources that are provisioned as services

Cloud stakeholders





Cloud service providers

Provide and render on-demand services

- Pay-as-you-go utility computing services to cloud users(e.g., telecommunication companies, systems integrators, large organization process outsourcers

Internet service providers

- Offer services accessible over the internet to cloud users, service brokers, consumers, and resellers

The cloud users

The cloud users use provisioned services to develop personalized products and web services (e.g., software applications and websites) for the end users' consumption based on prior formal requirements specifications

The end users

- These are the **direct consumers** of the **products developed** by the **cloud users**

The other stakeholders of the cloud ecosystem

Pages: 43 - 45

Cloud Computing Characteristics

Cloud Computing Characteristics

Are categorized as either **technical** or **basic**

Basic Characteristics

- On-demand self-service
- Broad network access
- Measured service
- Multitenancy
- Reliability
- Cost effectiveness
- Customization

Basic Characteristics

- Efficient resource utilization
- Maintainability
- Collaboration
- Virtualization
- Green technology
- High performance
- Shared infrastructure
- Network Access

Technical Characteristics of Cloud Computing

Characteristics	Description
Large-scale computing resources	Cloud computing systems are often implemented in large scale to reduce costs , and providers can benefit from significant economies of scale
Shared resource pooling	Resource pooling forms the foundation for cloud scalability
Dynamic resource scheduling	Through software automation , cloud resources are provisioned dynamically based on the current demand requirements and ensure that the cloud service capability is being expanded or contracted based on demand
High scalability	Cloud computing architecture allows you to add new nodes and servers with no reconfiguration and remodification requirements to cloud software and infrastructure
Rapid elasticity	Computing resources can be rapidly and elastically provisioned and released based on the demand of the consumer

Advantages and Disadvantages of Cloud Computing

Advantages and Disadvantages of Cloud Computing

Advantages

- Economical
- Almost unlimited storage
- Backup and recovery
- Automatic software integration
- Easy access to information
- Quick deployment

Disadvantages

- **Technical issues**
- Prone to attack
- Security and privacy
- Lack of standards
- Continuously evolving
- Compliance concerns
- Service migration

Business drivers of cloud computing adoption

Business Drivers of Cloud Computing Adoption

- Cloud computing is a cost-reducing technology that allows small and medium-scale enterprises to contribute to economic growth by strengthening their pro-activeness and productivity

Business Drivers of Cloud Computing Adoption

Help to pursue new business opportunities	Drives improved customer relationship by providing a platform to locate and interact with potential customers over the internet
Upfront costs reduction	This reduces capital expenditure costs that include the cost of computer systems, physical data center and technical staff The organization only pays for the computing resources actually used and makes a few reasonable operational expenditures

Business Drivers of Cloud Computing Adoption

Potential improvement in business continuity

With **guaranteed available** and **efficient network connectivity**, **cloud** ensures seamless **availability of computing infrastructure** with its increased capacity for **improved disaster recovery**

Potential reduction in carbon footprint

Inevitable **provisioning of electricity** and **air conditioning** is **greatly eliminated** due to the **efficient use of computer resources** with limited demand for electricity and air conditioning

Business Drivers of Cloud Computing Adoption

Server costs	A server has the capability to support a number of workloads
Storage costs Network costs Backup and archive costs Disaster recovery costs Software maintenance costs Platform costs Support personnel costs	(p,68 – 69)

Summary

Cloud computing

Defined as a model for **enabling** ubiquitous, convenient, on-demand network access to a shared pool of configurable computing resources that can be rapidly provisioned and released with minimal management effort or service provider interaction (NIST)

Cloud Computing Characteristics

Defined as the distinctive **features** and **qualities** that **differentiate** such systems from other seemingly-related systems

Cloud Computing Ecosystem

- Defined as a **dynamic** and **complex** community of the cloud computing system components and the stakeholders.
- The **three primary stakeholders** of the cloud include the **end users**, the **cloud users** and the **cloud providers**

Summary

Five technical characteristics of Cloud Computing

- Large-scale computing resources
- Shared resource Pooling
- Dynamic resource scheduling
- High scalability
- Rapid elasticity

Advantages of Cloud Computing

- Economical
- Unlimited storage
- Backup and recovery
- Automatic software integration
- Easy access to information
- Quick deployment.

Disadvantages of Cloud Computing

- Technical issues
- Prone to attack
- Security and privacy
- Compliance concerns
- Service Migration

**Next Lesson: Cloud Computing Service and
Deployment Models**
